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United States Patent	12416531
Kind Code	B2
Date of Patent	September 16, 2025
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Temperature sensor, temperature detection device and image formation device

Abstract

A temperature sensor that sufficiently has heat insulation property and the pressing force to a temperature measurement object to a degree allowing a ceramic paper to be substituted, and that has a good responsiveness, and a temperature detection device and an image formation device that include the temperature sensor. The temperature sensor is disposed so as to maintain an abutting state with a temperature measurement object, the temperature sensor including: a thermosensitive element configured to detect the temperature of the temperature measurement object; a heat collection member configured to pressurize the temperature measurement object and to be thermally coupled with the thermosensitive element; and a holding member supporting the heat collection member and forming a space that faces the heat collection member.

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Appl. No.:	**17/905204**
Filed (or PCT Filed):	**December 17, 2020**
PCT No.:	**PCT/JP2020/047200**
PCT Pub. No.:	**WO2021/210218**
PCT Pub. Date:	October 21, 2021

Prior Publication Data

Document Identifier	Publication Date

Foreign Application Priority Data

JP	2020072858	Apr. 15, 2020

Publication Classification

Int. Cl.: **G01K7/24** (20060101); **G01K13/08** (20060101)

U.S. Cl.:

CPC **G01K7/24** (20130101); **G01K13/08** (20130101);

Field of Classification Search

CPC: G01K (7/24); G01K (13/08)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a National Stage application of PCT international application PCT/JP2020/047200, filed on Dec. 17, 2020 which claims priority to Japanese Patent Application No. 2020-072858, filed on Apr. 15, 2020, the contents of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) The present invention relates to a temperature sensor that detects the temperature of an object, a temperature detection device and an image formation device.

BACKGROUND ART

(3) There has been known a temperature detection device that is disposed so as to contact with a heater provided in a thermal fixing roller for controlling the temperature of the thermal fixing roller included in an image formation device such as a printer using an electrophotographic process, for example (Patent Literature 1, for example).

(4) The temperature detection device in Patent Literature 1 includes a temperature detection element, a sensor main body having an insert-molded conduction member to provide the conduction between a lead wire of the temperature detection element and a covered electric wire of a circuit portion, and a heat-resistant elastic body interposed between the sensor main body and the temperature detection element. The sensor main body is elastically supported by a support body through a coil spring. The temperature detection element is pressed to the heater by the elastic force of the heat-resistant elastic body. As the heat-resistant elastic body, a ceramic paper formed from fibers composed of an inorganic material is typically used.

CITATION LIST

Patent Literature

(5) Patent Literature 1: JP 2002-122489 A

SUMMARY OF INVENTION

Technical Problem

(6) In recent years, there has been a demand for a temperature sensor in which the ceramic paper is not used.

(7) The present invention has an object to provide a temperature sensor that has heat insulation property, the pressing force to a temperature measurement object and a good responsiveness, without the ceramic paper, and a temperature detection device and an image formation device that include the temperature sensor.

Solution to Problem

(8) The present invention is a temperature sensor that is disposed so as to maintain an abutting state with a temperature measurement object, the temperature sensor characterized by including: a thermosensitive element configured to detect the temperature of the temperature measurement object; a heat collection member configured to pressurize the temperature measurement object and to be thermally coupled with the thermosensitive element; and a holding member supporting the heat collection member and forming a space that faces the heat collection member.

(9) In the temperature sensor of the present invention, it is preferable that the heat collection member be a plate spring.

(10) In the temperature sensor of the present invention, it is preferable that the heat collection member include a main body portion where the thermosensitive element is disposed, and that an end portion of the main body portion be supported by the holding member, and the main body portion be pressurized to the temperature measurement object side by elastic force.

(11) In the temperature sensor of the present invention, it is preferable that the heat collection member include a pair of leg portions supported by the holding member, at both end portions of the

main body portion, and that the main body portion be pressurized to the temperature measurement object side by the elastic force of the leg portions.
(12) In the temperature sensor of the present invention, it is preferable that an element disposition portion for disposing the thermosensitive element be formed at the main body portion.
(13) In the temperature sensor of the present invention, it is preferable that the holding member include an element support portion supporting the thermosensitive element through the element disposition portion.
(14) In the temperature sensor of the present invention, it is preferable that the element disposition portion be formed in a concave shape at a part of the main body portion, and that the thermosensitive element be accommodated in the element disposition portion.
(15) In the temperature sensor of the present invention, it is preferable that the main body portion be formed in a substantially rectangular shape in planar view, and be supported by the holding member, at leg portions included in both end portions in a longitudinal direction, and that the element disposition portion be formed so as to extend in a short direction of the main body portion and to bend in an out-of-plane direction of the main body portion.
(16) In the temperature sensor of the present invention, it is preferable that the heat collection member include a plurality of positioning pieces that are inserted into the space.
(17) In the temperature sensor of the present invention, it is preferable that the holding member include a wall body forming the space, and a contact protrusion protruding from the wall body and contacting with the positioning pieces be formed on the wall body.
(18) In the temperature sensor of the present invention, it is preferable that the holding member include a wall body forming the space, that the heat collection member be supported by a distal end of a part of the wall body, and that the wall body have a lower height at a position where the heat collection member is supported, than the height at the other position.
(19) In the temperature sensor of the present invention, it is preferable that a first insulating material covering the heat collection member from the temperature measurement object side be disposed between the thermosensitive element and the temperature measurement object.
(20) In the temperature sensor of the present invention, it is preferable that the first insulating material be formed in a film shape.
(21) In the temperature sensor of the present invention, it is preferable that the thermosensitive element include a thermosensitive body having a resistance value that changes depending on temperature change, and a lead wire for electrically connecting the thermosensitive body to an external circuit, and that a second insulating material configured to insulate at least the lead wire of the thermosensitive element and the heat collection member be disposed between the thermosensitive element and the heat collection member.
(22) In the temperature sensor of the present invention, it is preferable that the second insulating material be formed in a film shape, and cover the heat collection member from the temperature measurement object side.
(23) In the temperature sensor of the present invention, it is preferable that the thermosensitive element include a thermosensitive body having a resistance value that changes depending on temperature change, and a pair of lead wires for electrically connecting the thermosensitive body to an external circuit, and that the pair of lead wires extend in one direction with respect to the thermosensitive element, and extend into the holding member through one side surface of the holding member.
(24) In the temperature sensor of the present invention, it is preferable that the thermosensitive element include a thermosensitive body having a resistance value that changes depending on temperature change, and a pair of lead wires for electrically connecting the thermosensitive body to an external circuit, and that the pair of lead wires extend in both directions with respect to the thermosensitive element, and extend into the holding member through both side surfaces of the holding member.

(25) Further, a temperature detection device of the present invention is characterized by including: the above-described temperature sensor; and a circuit portion electrically connected to the temperature sensor and configured to calculate the temperature of the temperature measurement object based on a signal from the temperature sensor.
(26) Further, an image formation device of the present invention that is an electrophotographic image formation device, the image formation device including: a fixing device configured to fix toner to a recording medium by heating and pressurizing; and the above-described temperature sensor configured to detect the temperature of a member included in the fixing device.
Advantageous Effect of Invention
(27) According to the present invention, since the heat collection member pressurizes the temperature measurement object and the heat collection member is thermally coupled with the thermosensitive element, the heat input to the heat collection member is quickly transferred to the thermosensitive element, due to the heat conduction from the pressurized temperature measurement object. Due to this heat collection action and the heat insulation action of the space that faces the heat collection member, it is possible to keep the heat in the thermosensitive element more sufficiently, and therefore it is possible to realize such a good responsiveness that the detection temperature of the thermosensitive element immediately follows the temperature fluctuation of the temperature measurement object without using the so-called ceramic paper.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view showing the external appearance of a temperature sensor according to an embodiment of the present invention.
(2) FIG. **2** is a plan view showing a thermosensitive element included in the temperature sensor shown in FIG. **1**.
(3) FIG. **3** is a side view showing a holding member and lead wires of the thermosensitive element as viewed in the direction of an arrow III in FIG. **1**.
(4) FIG. **4**A is a perspective view showing the holding member.
(5) FIG. **4**B is a plan view showing the holding member.
(6) FIG. **5**A is a perspective view showing the holding member and a plate spring, two-dot chain lines showing the thermosensitive element.
(7) FIG. **5**B is a plan view showing the holding member and a plate spring, two-dot chain lines showing the thermosensitive element.
(8) FIG. **6**A is a perspective view showing the plate spring as a heat collection member.
(9) FIG. **6**B is a plan view showing the plate spring as a heat collection member.
(10) FIG. **7**A is a sectional view taken along a VIIa-VIIa line in FIG. **1**.
(11) FIG. **7**B is an enlarged view of a VIIb portion in FIG. **7**A.
(12) FIG. **8** is a perspective view showing a state where a film on the inside is provided in the holding member.
(13) FIG. **9** is a schematic view showing an internal structure of a printer equipped with the temperature sensor shown in FIG. **1**.
DESCRIPTION OF EMBODIMENT
(14) An embodiment (FIG. **1** to FIG. **8**) of the present invention will be described below with reference to the accompanying drawings.
(15) [Schematic Configurations of Temperature Detection Device and Temperature Sensor]
(16) Schematic configurations of a temperature detection device **1** and a temperature sensor **10** of the present invention will be described with reference to FIG. **1** and FIG. **7**A. As shown in FIG. **1**, the temperature detection device **1** is configured to include the temperature sensor **10**, a circuit

portion **8**, and electric wires **81**, **82** for electrically connecting the temperature sensor **10** and the circuit portion **8**.
(17) The temperature sensor **10** is disposed at a position facing a temperature measurement object **7** (see FIG. **7**A), so as to maintain an abutting state with the temperature measurement object **7**. As main constituent elements, the temperature sensor **10** includes a thermosensitive element **11** that detects the temperature of the temperature measurement object **7**, a holding member **20**, and a plate spring **30** as a heat collection member.
(18) Further, for securing insulation and creepage distance, the temperature sensor **10** includes an inside film **41** (second insulating material) that covers the plate spring **30** and an outside film **42** (first insulating material) that covers the thermosensitive element **11** disposed on the inside film **41**. Further, the temperature sensor **10** includes a heat collection material **43** that is encapsulated in the periphery of the thermosensitive element **11** between the inside film **41** and the outside film **42**.
(19) The circuit portion **8** calculates the temperature of the temperature measurement object **7**, based on an electrical signal that is output from the thermosensitive element **11**. The circuit portion **8** is electrically connected to the temperature sensor **10** through the electric wires **81**, **82** that are drawn out from the holding member **20**.
(20) A direction in which the temperature sensor **10** extends in a direction in which the electric wires **81**, **82** are drawn out is referred to as a longitudinal direction D**1**. A direction orthogonal to the longitudinal direction D**1** in planar view is referred to as a width direction D**2**. A direction orthogonal to both of the longitudinal direction D**1** and the width direction D**2** is referred to as a height direction D**3**. The temperature measurement object **7** side in the height direction D**3** is referred to as an “upper”, and the opposite side is referred to as a “lower”.
(21) Further, with respect to the position of the plate spring **30**, a front surface side Fs is defined as the temperature measurement object **7** side, and a back surface side Bs is defined as the opposite side (holding member **20** side). In the embodiment, the front surface side Fs corresponds to the upper side, and the back surface side Bs corresponds to the lower side.
(22) Each constituent element of the temperature sensor **10** will be described below.
(23) [Thermosensitive Element]
(24) As shown in FIG. **2**, the thermosensitive element **11** is a thermistor element that includes a thermosensitive body **110**, electrodes **110**A, **110**B provided in the thermosensitive body **110**, a pair of lead wires **111**, **112** electrically connected to the electrodes **110**A, **110**B, and a covering portion **113** covering the thermosensitive body **110**. In addition, as the thermosensitive element **11**, resistors having temperature coefficients, as exemplified by a thin film thermistor and a platinum temperature sensor, can be widely used.
(25) The lead wires **111**, **112** are respectively conducted to the electric wires **81**, **82** through a pair of conductive members **121**, **122** (FIG. **7**A) described later, which are provided in the holding member **20**.
(26) The thermosensitive body **110** of the thermosensitive element **11** is disposed so as to face the temperature measurement object **7** (FIG. **7**A) through the heat collection material **43** and the outside film **42**.
(27) [Holding Member]
(28) The holding member **20** will be described with reference to FIG. **4**A and FIG. **4**B.
(29) The holding member **20** in the embodiment is formed in a substantially rectangular shape in planar view, and includes a main body portion **22**, a seat **201** including a wall body **21** that is formed so as to protrude in the height direction D**3** from a nearly central portion of the main body portion **22** in the longitudinal direction D**1**, and an electric wire connection portion **25** to which the electric wires **81**, **82** are connected. A space **20**S surrounded by the wall body **21** and having a rectangular parallelepiped shape is formed on the inside of the seat **201**. The plate spring **30** described later is arranged on the front surface side Fs (the upper side in the height direction D**3**) of the space **20**S.

(30) The holding member **20** may be formed in a square shape or a circular shape in planar view, depending on the shape of the space **20**S, the disposition of bosses **221**, **231** described later, and the like.
(31) The holding member **20** is integrally formed by injection molding, using an insulating resin material. For example, an upper surface **22***a* of the main body portion **22** and an upper surface **25***a* of the electric wire connection portion **25** exist on an identical plane. Moreover, a plurality of first bosses **221** formed so as to protrude in the height direction D**3** are provided on the upper surface **22***a* of the main body portion **22**, and a plurality of second bosses **231** formed so as to protrude in the width direction D**2** are provided on a side surface **23** of the main body portion **22**. The first bosses **221** and the second bosses **231** are used for attaching the films **41**, **42** to the holding member **20**.
(32) The electric wire connection portion **25** is a site for attaching the electric wires **81**, **82** for electrically connecting the thermosensitive element **11** and the circuit portion **8**, and is formed integrally with the main body portion **22** at one end portion of the main body portion **22** in the longitudinal direction D**1**. Connection holes **251**, **252** for respectively connecting the electric wires **81**, **82** to the conductive members **121**, **122** described later are formed on the electric wire connection portion **25**.
(33) The seat **201** is a site for attaching the plate spring **30**, and is formed in a rectangular shape in planar view. The seat **201** is constituted by the wall body **21** and the space **20**S.
(34) The wall body **21** includes a pair of first walls **211**, **211** that extend in the longitudinal direction D**1** and that face each other in the width direction D**2**, and second walls **212**, **212** that connect the respective both ends of the first walls **211**, **211** in the longitudinal direction D**1**. Each of the first walls **211**, **211** and the second walls **212**, **212** is formed so as to stand in the height direction D**3** from a bottom portion **213** of the seat **201**, and upper ends of the first walls **211**, **211** and the second walls **212**, **212** form an opening **210** having a rectangular shape. Cutouts **211**A, **211**A formed so as to be cut in a concave shape are respectively formed at the centers of the first walls **211**, **211** in the longitudinal direction D**1**. Moreover, positioning pieces **321** to **324** (FIG. **6**) each of which is a part of the plate spring **30** are inserted into the space **20**S through the opening **210**.
(35) The space **20**S maintains heat in the thermosensitive element **11**, due to the heat insulation action that restrains the heat conduction from the thermosensitive element **11** to the exterior. Thereby, the resistance value of the thermosensitive element **11** is quickly changed in response to the temperature fluctuation of the temperature measurement object **7** that inputs heat to the thermosensitive element **11**, and the responsiveness of the temperature sensor **10** is enhanced. Cross-sectional areas (areas in the directions D**1** and D**2**) and a thickness (a size in the direction D**3**) that realize a necessary thermal resistance are given to the space **20**S.
(36) For keeping the heat conductivity of the space **20**S as low as possible, it is preferable that substances other than air be not disposed in the space **20**S. The existence of substances such as gas and liquid other than air is not entirely excluded in the space **20**S, and substances other than air is not avoided from being encapsulated into the space **20**S, as long as a low heat conductivity can be maintained. Further, it is allowable to dispose a member having a plate shape or the like in the space **20**S, for restraining the generation of a convective flow in the space **20**S. The space **20**S may be formed in another shape, for example, in a cylindrical shape.
(37) Support portions **214** supporting the plate spring **30** are formed on an upper end (distal end) of the second wall **212**. The support portions **214** are formed so as to be flat, and the height of the support portions **214** is set so as to be lower than the height of the upper end of the first wall **211**.
(38) Outer surfaces **212**A of the pair of the second walls **212** are inclined in such directions that the outer surfaces **212**A come close to each other upward. Therefore, the holding member **20** is formed in a frustum shape in side view.
(39) Element support portions **215** supporting the thermosensitive element **11** through the plate

spring **30**, first contact protrusions **216** and second contact protrusions **217** are formed on the inside of the wall body **21**.
(40) The element support portions **215** are formed so as to respectively protrude to the inside in the width direction D**2** from positions where the cutouts **211**A, **211**A of the pair of the first walls **211** are formed. Upper end surfaces of the element support portions **215** are continuous with bottom surfaces of the cutouts **211**A formed on the first walls **211**. Each element support portion **215** is positioned at the center of the space **20**S in the longitudinal direction D**1**, and supports the plate spring **30** at a position corresponding to the thermosensitive body **110**.
(41) The first contact protrusion **216** (FIG. **4**A and FIG. **4**B) is formed at two spots on both sides of the element support portions **215** of each of the pair of the first walls **211**. The respective first contact protrusions **216** protrude from the first walls **211**, and distal ends of the first contact protrusions **216** contact with surfaces of the positioning pieces **321** to **324** of the heat collection member **30**. Further, the second contact protrusions **217** are formed at four corners of the bottom portion **213** in the vicinity of the first contact protrusions **216**, and distal ends of the second contact protrusions **217** contact with side surfaces of the positioning pieces **321** to **324**. With the first contact protrusions **216** and the second contact protrusions **217**, the positioning of the plate spring **30** in the longitudinal direction D**1** and the width direction D**2** is performed, and the plate spring **30** is maintained in a state of being away from the wall body **21** as much as possible.
(42) As shown in FIG. **4**B, the first contact protrusions **216** are formed so as to protrude from the first walls **211** in a circular arc shape. Further, upper ends of the first contact protrusions **216** have a taper shape for guiding the positioning pieces **321** to **324** at the time of the insertion of the plate spring **30**. The height of the first contact protrusions **216** is set so as to be lower at a distal end side than at a proximal end on the first wall **211** side. Similarly, upper ends of the second contact protrusions **217** are formed in a taper shape.
(43) By adopting the shapes of the first contact protrusions **216** and the second contact protrusions **217**, the contact area between the holding member **20** and the plate spring **30** is reduced, and therefore it is hard for heat to escape from the thermosensitive element **11** to the holding member **20** through the plate spring **30**.
(44) As shown in FIG. **7**A, the conductive members **121**, **122** having a plate shape are provided in the bottom portion **213** of the seat **201**. The insert molding can be performed while the conductive members **121**, **122** are disposed in a mold for the injection molding of the holding member **20**. Connection holes **241**, **242** are formed so as to penetrate the bottom portion **213** in the height direction D**3**. The conductive members **121**, **122** are disposed so as to protrude to the insides of the connection holes **241**, **242**. The lead wires **111**, **112** (FIG. **3**) extend to one side of the plate spring **30** in the width direction D**2**, and extend to the inside of the bottom portion **213** through the side surface **23** of the holding member **20**.
(45) The first bosses **221** and the second bosses **231** are formed integrally with the holding member **20**.
(46) The first bosses **221** are disposed on the upper surface of the main body portion **22** such that two first bosses **221** face two first bosses **221** in the longitudinal direction D**1** through the seat **201**, and the second bosses **231** are disposed such that two second bosses **231** are disposed on each of both side surfaces **23** of the holding member **20**. This is an example, and the first bosses **221** and the second bosses **231** can be formed at appropriate positions of the holding member **20**.
(47) [Plate Spring (Heat Collection Member)]
(48) The plate spring **30** (FIG. **5**A to FIG. **7**B) pressurizes the temperature measurement object **7** from the back surface side Bs by elastic force, and is thermally coupled with the thermosensitive element **11**.
(49) For quickly transferring the heat from the temperature measurement object **7** to the thermosensitive element **11**, the plate spring **30** can be integrally formed using a metal material generally having a higher heat conductivity than resin material and the like, or another material

having a heat conductivity that is comparable to the heat conductivity of the metal material, for example, using a metal material such as a copper alloy and stainless steel, a material containing carbon, or the like. In the case where the metal material is used, the plate spring **30** can be formed by press working such as punching and bending. The material that is used for the plate spring **30** can be appropriately selected in consideration of heat conductivity, spring property and heat resistance property.
(50) It is preferable that the plate thickness of the plate spring **30** be as small as possible, as long as the strength can be secured, for avoiding the decrease in responsiveness due to the increase in heat capacity. For example, the plate thickness of the plate spring **30** is about 0.05 to 0.2 mm.
(51) Due to not only the heat insulation action of the above-described space **20**S but also the heat collection action of the plate spring **30**, it is possible to quickly change the resistance value of the thermosensitive element **11** in response to the temperature fluctuation of the temperature measurement object **7**, and to further enhance the responsiveness of the temperature sensor **10**. The "heat collection" in the specification means that the heat input from the temperature measurement object **7** is quickly transferred to the thermosensitive element **11**. With this heat collection action of the plate spring **30**, the heat is maintained in the thermosensitive element **11** and the vicinity of the thermosensitive element **11**.
(52) The plate spring **30** includes a main body portion **31** on which the thermosensitive element **11** is disposed and the plurality of positioning pieces **321** to **324** for the positioning of the plate spring **30** on the holding member **20**.
(53) The main body portion **31** is formed in a substantially rectangular shape in planar view. The main body portion **31** includes an abutting portion **310** that is pressed against the temperature measurement object **7**, and spring pieces **311**, **312** as a pair of leg portions that form both end portions of the abutting portion **310** in the longitudinal direction D**1**.
(54) A groove **310**A as an element disposition portion where the thermosensitive body **110** and parts of the lead wires **111**, **112** are disposed is provided on the abutting portion **310**. The groove **310**A extends in the short direction (width direction D**2**) of the main body portion **31** and is bent in an out-of-plane direction of the main body portion **31** to form a concave shape. The groove **310**A is formed from one end of the abutting portion **310** to the other end of the abutting portion **310** in the width direction D**2**.
(55) The groove **310**A is set so as to have a width (the size in the longitudinal direction D**1**) and a depth that allow the thermosensitive body **110** to be accommodated even when the sizing of the thermosensitive body **110** varies due to tolerance.
(56) When heat is sufficiently input from the temperature measurement object **7** to the lead wires **111**, **112** (particularly, Dumet wires in the vicinity of the thermosensitive body **110**) having a higher heat conductivity than the thermosensitive body **110**, the responsiveness can be enhanced, and therefore it is preferable that sections of the lead wires **111**, **112** that are disposed in the groove **310**A be as long as possible, in consideration of the position where the thermosensitive body **110** of the thermosensitive element **11** is disposed. For example, the thermosensitive body **110** is disposed so as to be shifted to the outside of the center in the width direction D**2** of the groove **310**A formed on the plate spring **30**, and thereby it is possible to elongate the sections of the lead wires **111**, **112** that are disposed in the groove **310**A.
(57) The abutting portion **310** is formed so as to be substantially flat except the groove **310**A, and the whole in the length direction D**1** is pressed against the temperature measurement object **7**.
(58) In order to avoid the abutting portion **310** supported by the spring pieces **311**, **312** at both ends from warping between both ends due to reaction force from the temperature measurement object, thereby causing the thermosensitive element **11** to move away from the temperature measurement object **7**, the abutting portion **310** is supported at the position of the groove **310**A by the element support portion **215** of the holding member **20**.
(59) Instead of the groove **310**A, a concave portion having the same shape as the outer shape of the

thermosensitive body **110** may be formed on the abutting portion **310**.
(60) The spring pieces **311**, **312** protrude from the abutting portion **310** to both sides in the longitudinal direction D**1**, and are bent so as to be inclined downward with respect to the surface of the abutting portion **310**, and the respective distal ends **311**A, **312**A are bent in the surface direction of the abutting portion **310** so as to perform surface contact with the support portions **214**. Between the pair of first walls **211** of the holding member **20**, the spring pieces **311**, **312** are respectively supported by the support portions **214** formed at positions lower than the first walls **211**. Therefore, when the abutting portion **310** is pressed against the temperature measurement object **7** and the spring pieces **311**, **312** are displaced on the support portions **214**, the spring pieces **311**, **312** elastically deform without interfering with the second walls **212**, and therefore it is possible to sufficiently pressurize the temperature measurement object **7**, as the plate spring **30**.
(61) The respective distal ends **311**A, **312**A of the spring pieces **311**, **312** are formed so as to be bent and to perform surface contact with the support portions **214**, and therefore, when the spring pieces **311**, **312** slide on the support portions **214**, it is hard for the spring pieces **311**, **312** to damage the holding member **20**.
(62) It is preferable that the contact area between the plate spring **30** and the holding member **20** be as small as possible. This is because heat escapes to the holding member **20** through the portion of the contact. Therefore, the spring pieces **311**, **312** of the plate spring **30** in the embodiment are formed in a biforked shape for decreasing the contact area with the support portions **214**.
(63) The positioning pieces **321** to **324** are formed on both sides of the abutting portion **310** in the width direction D**2**. The positioning pieces **321** to **324** are respectively provided close to four corners of the abutting portion **310**, and are formed so as to protrude downward by bending the abutting portion **310** in the thickness direction D**3**. The positioning pieces **321** to **324** communicate with portions **313** of the abutting portion **310** where the width is decreased. In the abutting portion **310**, the width is decreased from steps **310**B, on both sides of the groove **310**A in the longitudinal direction D**1**.
(64) When the positioning pieces **321** to **324** are inserted into the space **20**S, the positioning is performed on the holding member **20** by the above-described first contact protrusions **216** and second contact protrusions **217**.
(65) When the temperature sensor **10** is pressed against the temperature measurement object **7** (FIG. **7**A), the abutting portion **310** performs surface contact with the temperature measurement object **7**, and the plate spring **30** is pressed by the temperature measurement object **7**. Then, the spring pieces **311**, **312** of the plate spring **30** elastically deform while being displaced on the support portions **214** to the outsides in the longitudinal direction D**1** orthogonal to the pressing direction (D**3**). At this time, the main body portion **31** is pressurized to the temperature measurement object **7** by the elastic force of the spring pieces **311**, **312**.
(66) [Inside Film]
(67) The inside film **41** is provided on the holding member **20**, for insulating the plate spring **30** and the thermosensitive element **11**.
(68) The inside film **41**, which has insulation property, is inserted into the inside of the groove **310**A, and is interposed between the plate spring **30** and the thermosensitive element **11**, as shown in FIG. **7**B and FIG. **8**. In addition, for sufficiently securing the creepage distance from the thermosensitive element **11**, the inside film **41** covers the whole of the surface of the plate spring **30** from the temperature measurement object **7** side.
(69) For the inside film **41**, for example, a resin material such as polyimide and a fluorocarbon resin is used. For example, the thickness of the inside film **41** is about 10 to 20 μm.
(70) The inside film **41** has a width equivalent to the width of the seat **201**, and is formed in a rectangular shape.
(71) In the case where insulating coatings are provided on the lead wires **111**, **112**, the inside film **41** is not always necessary even when the plate spring **30** has conductive property. Similarly, in the

case where the plate spring **30** is a member having no conductive property, as exemplified by a resin molded article, it is possible to omit the arrangement of the inside film **41**.

(72) Moreover, the plate spring **30** and the thermosensitive element **11** are thermally coupled through the inside film **41**.

(73) [Heat Collection Material]

(74) For collecting heat in the thermosensitive element **11**, it is preferable that a gap G**1** (FIG. **7**B) in the periphery of the thermosensitive element **11** disposed in the groove **310**A be filled with the insulating heat collection material **43** (FIG. **7**B and FIG. **8**) that is thermally coupled with the thermosensitive element **11**.

(75) As the heat collection material **43**, for example, a material containing a disperse medium such as a silicone resin that has a high heat conductivity among resin materials, and an insulating dispersoid such as a ceramic powder is used. Further, as the heat collection material **43**, a so-called heat conductive grease or silicone oil compound can be used.

(76) The filling with the heat collection material **43** is performed on the inside film **41**.

(77) The heat collection material **43** tightly contacts with the thermosensitive element **11** and tightly contacts with the plate spring **30** through the inside film **41**, and thereby it is possible to thermally couple the plate spring **30** with the thermosensitive element **11** more sufficiently, and to further enhance the responsiveness.

(78) [Outside Film]

(79) The outside film **42** has insulation property, and is provided for insulating the thermosensitive element **11** and the temperature measurement object **7** by covering the plate spring **30** and the holding member **20** from the temperature measurement object **7** side while being interposed between the thermosensitive element **11** and the temperature measurement object **7**. Further, the outside film **42** holds the thermosensitive element **11** on the plate spring **30**. As shown in FIG. **1** and FIG. **7**A, the outside film **42** covers the thermosensitive element **11** on the inside film **41**, and is fixed to both side surfaces **23** of the holding member **20**. Furthermore, for sufficiently securing the creepage distance from the thermosensitive element **11** and the conductive members **121**, **122**, the outside film **42** covers most of the holding member **20** that includes both side surfaces **23** of the holding member **20**, in addition to the whole of the plate spring **30**.

(80) For the outside film **42**, for example, a resin material such as polyimide and a fluorocarbon resin is used. As shown in FIG. **7**B, the outside film **42** may be a film in which two or more film materials are laminated. For example, the whole thickness of the outside film **42** is about 10 to 20 μm.

(81) [Assembly Procedure for Temperature Sensor]

(82) For example, the temperature sensor **10** can be assembled in the following procedure.

(83) Each procedure will be described.

(84) (1) Terminals of the lead wires **111**, **112** of the thermosensitive element **11** are joined to the conductive members **121**, **122** (FIG. **7**A) in a state where the insert molding has been performed in the holding member **20**, by electric welding or the like.

(85) (2) The positioning pieces **321** to **324** of the plate spring **30** are inserted into the inside of the wall body **21** of the holding member **20** shown in FIG. **4**A and FIG. **4**B. At this time, distal ends (lower ends) of the positioning pieces **321** to **324** do not contact with the bottom portion **213**. When the plate spring **30** is further pushed to the inside of the wall body **21**, the distal ends of the positioning pieces **321** to **324** come in contact with the bottom portion **213**, and the positioning pieces **321** to **324** function as a stopper at this time.

(86) As shown in FIG. **5**A, the plate spring **30** is assembled to the holding member **20** in a state where the spring pieces **311**, **312** are supported by the support portions **214**. At this time, the abutting portion **310** protrudes upward of the upper ends of the first walls **211**. The space **20**S is defined on the back surface side Bs of the plate spring **30**.

(87) (3) Next, as shown in FIG. **8**, the plate spring **30** is covered by the inside film **41**, and the

inside film **41** is fixed to the holding member **20** by a heat caulking in which heat and pressure are applied to the first bosses **221** that are inserted into holes provided at four corners of the inside film **41**.
(88) (4) The lead wires **111**, **112** are routed while being shaped, and the thermosensitive element **11** is assembled to the holding member **20** and the plate spring **30**. Specifically, the lead wires **111**, **112** are bent upward along the side surfaces **23** of the holding member **20** from the conductive members **121**, **122**, and thereafter are also bent at the position of the groove **310**A of the plate spring **30**. Then, the thermosensitive body **110** is accommodated in the groove **310**A on which the inside film **41** is laid.
(89) (5) Subsequently, the heat collection material **43** is applied to the thermosensitive body **110**. The heat collection material **43** adheres to the thermosensitive body **110** and the lead wires **111**, **112** drawn out from the thermosensitive body **110**.
(90) (6) Finally, as shown in FIG. **1**, most of the holding member **20** that includes the thermosensitive element **11** and the heat collection material **43** is covered by the outside film **42**, and the outside film **42** is fixed to the holding member **20** by the heat caulking of the second bosses **231**, similarly to the first bosses **221**.
(91) Thereby, the assembly of the temperature sensor **10** is completed.
(92) [Main Function Effect of Temperature Sensor]
(93) When the temperature sensor **10** is installed so as to face the temperature measurement object **7** (FIG. **7**A), the abutting portion **310** of the plate spring **30** is pressed against the temperature measurement object **7**, and thereby the spring pieces **311**, **312** elastically deform while smoothly sliding on the support portions **214**. As a result, the abutting portion **310** supported by the element support portions **215** is evenly pressurized to the temperature measurement object **7** through the inside film **41** and the outside film **42**. Furthermore, the heat radiated from the temperature measurement object **7** is sufficiently input to the abutting portion **310** pressed against the temperature measurement object **7**, and the heat is sufficiently transferred from the abutting portion **310** to the thermosensitive element **11** through the heat collection material **43** supplied to the inside of the groove **310**A. At this time, the abutting portion **310** is thermally coupled with not only the thermosensitive element **11** but also the temperature measurement object **7**.
(94) Even when void remains between the thermosensitive element **11** and the abutting portion **310** at the time of the completion of the assembly of the temperature sensor **10**, the thermosensitive body **110** and the vicinity of the thermosensitive body **110** are filled with the heat collection material **43** substantially with no gap, by the pressurization from the plate spring **30** to the temperature measurement object **7**. Therefore, it is possible to thermally couple the thermosensitive element **11** and the plate spring **30** more sufficiently.
(95) Furthermore, the positioning pieces **321** to **324** and spring pieces **311**, **312** of the plate spring **30** contact with the holding member **20** by the minimum necessary contact, and therefore the outflow of the heat to the holding member **20** through the plate spring **30** is restrained. By the synergy of this, the heat collection action of the plate spring **30**, and further the heat insulation action of the space **20**S having a lower heat conductivity than a heat insulating material such as a ceramic paper, it is possible to more surely keep the heat transferred from the temperature measurement object **7** in the thermosensitive element **11** and the vicinity of the thermosensitive element **11**.
(96) Accordingly, with the temperature sensor **10** in the embodiment, it is possible to cause the thermosensitive element **11** to immediately follow the temperature change of the temperature measurement object **7**, and to realize a good responsiveness of the detection temperature. Due to the thin plate spring **30** and the space **20**S defined on the back surface side Bs, the temperature sensor **10** has both the heat insulation performance and the elastic force for pressurizing the temperature sensor **10** to the temperature measurement object **7**, and therefore without using the so-called ceramic paper, it is possible to realize a responsiveness equal to or higher than that of the

ceramic paper, while maintaining the small size of the temperature sensor **10**.
(97) [Application Example for Image Formation Device]
(98) An example in which the temperature detection device **1** including the temperature sensor **10** is applied to the laser printer **9** as an example of the image formation device will be briefly described with reference to FIG. **9**.
(99) As shown in FIG. **9**, the laser printer **9** includes a photoreceptor belt **91**, an electrifier **92**, an exposure device **93**, developing devices **901** to **904**, a guide roller **94**, an intermediate transfer unit **95**, a paper feed cassette **96**, a paper feed roller **97**, a transfer roller **98**, a fixing device **99**, a resist roller **910**, a paper ejection roller **911**, a paper ejection tray **912**, and a control device **900** that controls each unit of the laser printer **9**.
(100) The fixing device **99** includes a pressure roller **991** and a heating roller **992**. The heating roller **992** incorporates an unillustrated heater as a heat source.
(101) For measuring the temperature of the heater incorporated in the heating roller **992** and the temperature of a member provided on the heater, the temperature sensor **10** is installed so as to be pressed against the heater or the member.
(102) After electrification, exposure, development and transfer as processes of the image formation by the laser printer **9**, a recording paper **913** on which a color toner image is transferred is fed to between the pressure roller **991** and heating roller **992** of the fixing device **99**, in the process of fixation. The recording paper **913** is pressurized and heated while passing through between the pressure roller **991** and the heating roller **992**, and thereby the color toner image is fixed to the recording paper **913**. Thereafter, the recording paper **913** is ejected to the paper ejection tray **912** through the paper ejection roller **911**.
(103) The control device **900** controls the energization state of the heater of the heating roller **992**, using the temperature measurement value obtained from the temperature sensor **10** and the circuit portion **8** to which the temperature sensor **10** is connected. For example, when the temperature measurement value exceeds a threshold, the control device **900** stops the energization of the heater of the heating roller **992**.
(104) With the temperature sensor **10**, the surface temperature of the heating roller **992** is measured at a high following capability, and therefore, it is possible to adequately control the energization state of the heater without needlessly heating the heating roller **992** with the heater in prospect of the delay of the response of the measurement.
(105) In addition to the above description, configurations described in the above embodiment can be chosen, or can be appropriately altered to other configurations, without departing from the spirit of the present invention.
(106) The case of using the thermistor element having a configuration in which the lead wires **111**, **112** extend from one side of the thermosensitive body **110** in one direction as shown in FIG. **2** has been exemplified and described in the above embodiment, but a thermistor element in which the lead wires **111**, **112** extend in both directions of the thermosensitive body **110** may be used. In this case, the lead wires **111**, **112** pass through both side surfaces **23** of the holding member **20**, and extend into the holding member **20**.
(107) Further, the case where the lead wires **111**, **112** are drawn out from the thermosensitive body **110** in the width direction D**2** has been exemplified and described in the embodiment, but the lead wires **111**, **112** may be drawn out from the thermosensitive body **110** in the longitudinal direction D**1**.
(108) The shape of the plate spring **30** is not limited to the above embodiment, and various modifications can be made.
(109) For example, a protrusion for performing the positioning of the plate spring **30** on the holding member **20** can be formed between portions of the biforked shape of each of the spring pieces **311**, **312**. In that case, the positioning pieces **321** to **324** extending downward from the abutting portion **310** is not necessary.

REFERENCE SIGNS LIST

(110) **1** temperature detection device **7** temperature measurement object **8** circuit portion **9** laser printer **10** temperature sensor **11** thermosensitive element **20** holding member **20***s* space **21** wall body **22** main body portion **23** side surface **25** electric wire connection portion **30** plate spring (heat collection member) **31** main body portion **41** inside film (second insulating material) **42** outside film (first insulating material) **43** heat collection material **81**, **82** electric wire **91** photoreceptor belt **92** electrifier **93** exposure device **94** guide roller **95** intermediate transfer unit **96** paper feed cassette **97** paper feed roller **98** transfer roller **99** fixing device **110** thermosensitive body **111**, **112** lead wire **113** covering portion **121**, **122** conductive member **201** seat **210** opening **211** first wall **211**A cutout **212** second wall **212**A outer surface **213** bottom portion **214** support portion **215** element support portion **216** first contact protrusion (contact protrusion) **217** second contact protrusion (contact protrusion) **221** first boss **231** second boss **241**, **242** connection hole **251**, **252** connection hole **310** abutting portion **310**A groove (element disposition portion) **310**B step **311**, **312** spring piece (leg portion) **311**A, **312**A distal end **321** to **324** positioning piece **900** control device **901** to **904** developing device **910** resist roller **911** paper ejection roller **912** paper ejection tray **913** recording paper (recording medium) **991** pressure roller **992** heating roller Bs back surface side Fs front surface side D**1** longitudinal direction D**2** width direction D**3** height direction G**1** gap

Claims

1. A temperature sensor that is disposed so as to maintain an abutting state with a temperature measurement object, the temperature sensor comprising: a thermosensitive element configured to detect a temperature of the temperature measurement object; a heat collection member configured to pressurize the temperature measurement object and to be thermally coupled with the thermosensitive element; and a holding member supporting the heat collection member and forming a space that faces the heat collection member, wherein the heat collection member is a plate spring.
2. The temperature sensor according to claim 1, wherein: the heat collection member includes a main body portion where the thermosensitive element is disposed; and an end portion of the main body portion is supported by the holding member, and the main body portion is pressurized to a temperature measurement object side by elastic force.
3. The temperature sensor according to claim 2, wherein: the heat collection member includes a pair of leg portions supported by the holding member, at both end portions of the main body portion; and the main body portion is pressurized to the temperature measurement object side by the elastic force of the leg portions.
4. The temperature sensor according to claim 2, wherein an element disposition portion for disposing the thermosensitive element is formed at the main body portion.
5. The temperature sensor according to claim 4, wherein the holding member includes an element support portion supporting the thermosensitive element through the element disposition portion.
6. The temperature sensor according to claim 4, wherein: the element disposition portion is formed in a concave shape at a part of the main body portion; and the thermosensitive element is accommodated in the element disposition portion.
7. The temperature sensor according to claim 6, wherein: the main body portion is formed in a substantially rectangular shape in planar view, and is supported by the holding member, at leg portions included in both end portions in a longitudinal direction; and the element disposition portion is formed so as to extend in a short direction of the main body portion and to bend in an out-of-plane direction of the main body portion.
8. The temperature sensor according to claim 1, wherein the heat collection member includes a plurality of positioning pieces that are inserted into the space.
9. The temperature sensor according to claim 8, wherein: the holding member includes a wall body

forming the space; and a contact protrusion protruding from the wall body and contacting with the positioning pieces is formed on the wall body.

10. The temperature sensor according to claim 1, wherein: the holding member includes a wall body forming the space; the heat collection member is supported by a distal end of a part of the wall body; and the wall body has a lower height at a position where the heat collection member is supported, than a height at the other position.

11. The temperature sensor according to claim 1, wherein a first insulating material covering the heat collection member from a temperature measurement object side is disposed between the thermosensitive element and the temperature measurement object.

12. The temperature sensor according to claim 11, wherein the first insulating material is formed in a film shape.

13. The temperature sensor according to claim 1, wherein: the thermosensitive element includes a thermosensitive body having a resistance value that changes depending on temperature change, and a lead wire for electrically connecting the thermosensitive body to an external circuit; and a second insulating material configured to insulate at least the lead wire of the thermosensitive element and the heat collection member is disposed between the thermosensitive element and the heat collection member.

14. The temperature sensor according to claim 13, wherein the second insulating material is formed in a film shape, and covers the heat collection member from a temperature measurement object side.

15. The temperature sensor according to claim 1, wherein: the thermosensitive element includes a thermosensitive body having a resistance value that changes depending on temperature change, and a pair of lead wires for electrically connecting the thermosensitive body to an external circuit; and the pair of lead wires extend in one direction with respect to the thermosensitive element, and extend into the holding member through one side surface of the holding member.

16. The temperature sensor according to claim 1, wherein: the thermosensitive element includes a thermosensitive body having a resistance value that changes depending on temperature change, and a pair of lead wires for electrically connecting the thermosensitive body to an external circuit; and the pair of lead wires extend in both directions with respect to the thermosensitive element, and extend into the holding member through both side surfaces of the holding member.

17. A temperature detection device comprising: the temperature sensor according to claim 1; and a circuit portion electrically connected to the temperature sensor and configured to calculate a temperature of the temperature measurement object based on a signal from the temperature sensor.

18. An image formation device that is an electrophotographic image formation device, the image formation device comprising: a fixing device configured to fix toner to a recording medium by heating and pressurizing; and the temperature sensor according to claim 1, the temperature sensor being configured to detect a temperature of a member included in the fixing device.